

Monday, January 27, 2020 1:05 PM

1) a) $A + B \rightleftharpoons C$

$$K_c = \frac{k_f}{k_b} \Rightarrow k_b = \frac{k_f}{K_c}$$

$$-\mu_A = k_f C_A C_B - k_b C_C$$

$$= k_f C_A C_B - \frac{k_f}{K_c} C_C$$

$$= k \left(C_A C_B - \frac{C_C}{K_c} \right) \quad (k = k_f)$$

$$C_A = C_{A0}(1-X) = C_B ; C_c = C_{A0}X$$

At equilibrium, $r_A = 0$

$$\Rightarrow C_A C_B = \frac{C_c}{K_c}$$

$$\Rightarrow C_{A_0}^2 (1 - x_e)^2 \cdot K_c = C_{A_0} x_e$$

$$\Rightarrow C_A(1 + X_e^2 - 2X_e)K_c - X_e = 0$$

$$\Rightarrow X_e^2 - 2X_e + 1 - \frac{X_e}{K_c C_M} = 0$$

$$\Rightarrow X_e^2 - X_e \left(2 + \frac{1}{K_G} \right) + 1 = 0$$

$$\Rightarrow X_e = 0.8$$

$$C_A = C_B = C_{A_0} (1 - X_e) = 0,4 \text{ M}$$

$$C_c = C_{A_b} X = 1.6 \text{ M.}$$

b) $A \rightarrow 3C$

$$V = V_0 + 2N_0X = V_0(1 + 2X)$$

$$N_A = N_{A_0} (1 - x)$$

$$N_e = N_{A_0} 3^X$$

$$C_A = \frac{N_A}{V} = \frac{N_{A0}(1-X)}{V(1+2X)} = C_{A0} \frac{(1-X)}{(1+2X)}$$

$$C_c = \frac{N_c}{V} = \frac{N_{A_0} 3x}{V(1+2x)} = C_{A_0} \frac{3x}{1+2x}$$

$$\bullet \quad K_c = \frac{C_c^3}{C_A} \Rightarrow C_c^3 = K_c C_A$$

$$\Rightarrow K_c \cdot G_0 \frac{(1-x)}{(1+2x)} = \left[G_0 \frac{3x}{(1+2x)} \right]^3$$

$$\Rightarrow K_c (1-x) = C_{A_0}^2 \frac{27x^3}{(1+2x)^2}$$

$$\Rightarrow (1+2x)^2 K_c (1-x) - C_{A_0}^2 27 x^3 = 0$$

$$\Rightarrow -\left(4 + \frac{27G^2}{4}\right)X_e^3 + 3X_e + 1 = 0$$

$$\Rightarrow X_e = 0.58.$$

$$\bullet C_A = \frac{P_A}{RT_A} = \frac{10 \text{ atm}}{(400 \text{ K}) \left(0.082 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}} \right)} = 0.305 \text{ M.}$$

$$C_A = 0.305 \frac{1 - 0.5^8}{1 + 2 \cdot 0.5^8} = 0.059 \text{ M}$$

$$C_c = 0.305 \frac{3 \times 0.58}{1.6 \times 10^3} = 0.246 \text{ M.}$$

2) $A \rightarrow B$ $k = 0.0015 \text{ min}^{-1}$ at 80°F .

$$M_A = M_B = 58 \text{ g/mol} \quad \text{activation energy} = 25,000 \text{ cal/mol.}$$

 $L = 10 \text{ ft}, d = 1 \text{ ft}$

$P = 132 \text{ psig}, 260^\circ\text{F.}$

$X_A = 0.9$, 1000 lb/hr of B $\rightarrow$ # tubes ?

$$\bullet F_B = \frac{1000 \frac{\text{lb}}{\text{hr}}}{58 \frac{\text{lb}}{\text{lb}}} = 17.21 \frac{\text{lb mol}}{\text{hr}}$$

$$\bullet F_B = F_A \times \Rightarrow F_A = \frac{F_B}{x} = \frac{17.21}{0.88} = 19.1 \frac{\text{lb} \cdot \text{mol}}{\text{h}}$$

- PFR:
$$V = \frac{n \pi d^2 L}{4L} = F_{A_0} \int_{-x_0}^{0.9} \frac{dx}{-r_x}$$

$$\bullet - r_A = k C_n = k C_A (1-x)$$

$$\bullet C_{A_0} = \frac{n}{V} = \frac{P}{RT}$$

$$\bullet V = F_{A_0} \int_0^{0.9} \frac{dX}{k_G(1-X)} = \frac{F_{A_0}}{k_G A_0} \int_0^{0.9} \frac{dX}{1-X}$$

$$= \frac{F_{A_0} RT}{k_p} \ln\left(\frac{1}{1-0.9}\right) = \frac{F_{A_0} RT}{k_p} \ln 10$$

$$\bullet \quad k_2 = k_1 \exp\left(\frac{E}{R} \left(\frac{1}{T_1} - \frac{1}{T_2}\right)\right)$$

$$= 0.0015 \exp\left(\frac{25000}{184} \left(\frac{1}{298} - \frac{1}{318}\right)\right) = 53.6 \text{ min}^{-1} = 3219 \text{ hr}^{-1}$$

$$T_1 = 80^\circ\text{F} = 540^\circ\text{R} ; T_2 = 260^\circ\text{F} = 720^\circ\text{R}$$

$$V = \frac{F_A \cdot R \cdot T}{R \cdot P} \ln 10 = \frac{(19.1 \frac{\text{lb} \cdot \text{mol}}{\text{hr}}) (10.73 \frac{\text{ft}^3 \cdot \text{psia}}{\text{lb} \cdot \text{mol} \cdot \text{R}}) (720 \text{ R})}{(3219 \text{ hr}^{-1}) (132 \text{ psia})} \ln 10$$

$$= 6.8 \text{ ft}^3$$

$$V = \frac{n\pi d^2 L}{4} \Rightarrow n = \frac{4V}{\pi d^2 L} = \frac{4 \times (0.72 \text{ ft}^3)}{\pi \left(\frac{1}{12} \text{ ft} \right)^2 (10 \text{ ft})} = 14.5$$

2. 15 July